

NOFT, and efficient approach for strongly-correlated systems

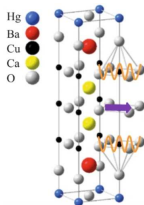
M. Rodríguez-Mayorga

Laboratoire de Physique Théorique - Toulouse (France)

Abinit Developer Workshop 2026
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Strongly correlated systems...

At normal (ambient) pressure $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+\delta}$ is superconductor at 138K (-135°C).



V. Oliviero et al., *Nature Communications*, **13**, 1568 (2022)

- Strongly-correlated systems tend to be unconventional superconductors... Not described by BCS theory!

But they are also semiconductors, Mott insulators, etc.

Need a computational method able to deal with strong correlation (beyond KS-DFT) and, ideally, fully *ab initio* (beyond DMFT)

The 1-RDM

The density matrix

$$n_1(\mathbf{r}, \mathbf{r}') = N \int d\mathbf{r}_2 \dots d\mathbf{r}_N \Psi^*(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N) \Psi(\mathbf{r}', \mathbf{r}_2, \dots, \mathbf{r}_N) \quad \rightarrow n(\mathbf{r}) = n_1(\mathbf{r}, \mathbf{r})$$

such that when we use a basis ϕ

$$n_1(\mathbf{r}, \mathbf{r}') = \sum_{pq} {}^1D_p^q \phi_p^*(\mathbf{r}) \phi_q(\mathbf{r}')$$

with ${}^1D_p^q = \langle \Psi | \hat{c}_p^\dagger \hat{c}_q | \Psi \rangle$

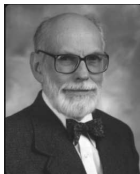
Since ${}^1\mathbf{D}$ is Hermitian we can diagonalize it

$$n_1(\mathbf{r}, \mathbf{r}') = \sum_p n_p \chi_p^*(\mathbf{r}) \chi_p(\mathbf{r}')$$

with $\sum_p n_p = N$, $n_p \in [0, 1]$ and χ are the natural orbitals.

Natural Orbital Funcional Theory (NOFT)

In 1975 T.L. Gilbert proposed reduced density matrix functional theory (a.k.a. NOFT) as an extension of DFT when $v_{\text{ext}}(\mathbf{r}, \mathbf{r}')$!



T.L. Gilbert, *PRB*, **12**, 2111 (1975)

When is $v_{\text{ext}}(\mathbf{r}, \mathbf{r}')$?

We we add the non-local term of a pseudo-potential, use magnetic fields, etc.

In this theory

$$E = E[n_1]$$

actually

$$E[n_1] = T[n_1] + V_{\text{ext}}[n_1] + V_{Hx}[n_1] + V_c[n_1]$$

only $V_c[n_1]$ is unknown! $\rightarrow$ Need approximations!

The NOFT functional for 2e⁻ systems

In 1956, P.-O. Löwdin and H. Shull analyzed the wavefunction for 2e⁻ systems in the natural orbital basis



P.-O. Löwdin and H. Shull, *PR*, **101**, 1730 (1956)

To find

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \sum_p c_p \chi_p(\mathbf{r}_1) \chi_p(\mathbf{r}_2) \quad \rightarrow \text{with } c_p = \pm \sqrt{n_p}$$

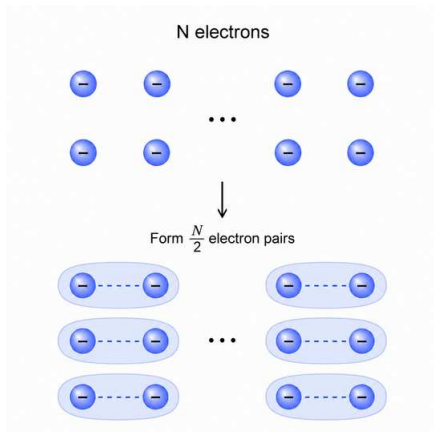
thus

$$E^{2e^-}[\mathbf{n}, \chi] = 2 \sum_p n_p h_{pp} + n_1 \langle 11 | 11 \rangle - 2 \sum_{p>1} \sqrt{n_1 n_p} \operatorname{Re} [\langle 11 | pp \rangle] + \sum_{p,q>1} \sqrt{n_p n_q} \langle pp | qq \rangle$$

with

$$h_{pq} = \int d\mathbf{r} d\mathbf{r}' \chi_p(\mathbf{r}) \left[-\delta(\mathbf{r} - \mathbf{r}') \frac{\nabla^2 \mathbf{r}'}{2} + v_{\text{ext}}(\mathbf{r}, \mathbf{r}') \right] \chi_q(\mathbf{r}') \text{ and } \langle pq | rs \rangle = \int d\mathbf{r} d\mathbf{r}' \frac{\chi_p^*(\mathbf{r}) \chi_q^*(\mathbf{r}') \chi_r(\mathbf{r}) \chi_s(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}$$

The NOFT functional for N e^- systems (as $N/2$ pairs)



Such that

$$E_{NOFT}[\mathbf{n}, \chi] = \sum_{\mu}^{N_{pairs}} E_{\mu}^{2e^{-}} + \sum_{\mu \neq \nu}^{N_{pairs}} E_{\mu\nu}^{inter}$$

The interaction energy



M. Piris, *Chem. Sci.*, **15**, 17284 (2024)

► PNOF5

$$E_{\mu\nu}^{inter} = \sum_{p \in \mu, q \in \nu} n_p n_q [2\langle pq|pq\rangle - \langle pq|qp\rangle]$$

This leads to a wavefunction method (Ψ_{APSG})!

► PNOF7

$$E_{\mu\nu}^{inter} = \sum_{p \in \mu, q \in \nu} n_p n_q [2\langle pq|pq\rangle - \langle pq|qp\rangle] \\ - \sum_{p \in \mu, q \in \nu} \sqrt{n_p(1-n_p)n_q(1-n_q)} \langle pp|qq\rangle$$

► PNOF8/GNOF

$$E_{\mu\nu}^{inter} = \sum_{p \in \mu, q \in \nu} n_p n_q [2\langle pq|pq\rangle - \langle pq|qp\rangle] + \sum_{p \in \mu, q \in \nu} f(n_p, n_q) \langle pp|qq\rangle$$

The optimization I

$$\mathcal{L}[\mathbf{n}, \chi, \lambda, \mu] = E_{NOFT}[\mathbf{n}, \chi] - \sum_{pq} \lambda_{pq} (\langle p|q \rangle - \delta_{pq}) - \mu \left(\sum_p n_p - N \right)$$

Optimization of $\mathbf{n}$ (with $n_p \in [0, 1]$)

Setting $n_p = \cos^2 \gamma_p$ and taking

$$\frac{\partial \mathcal{L}}{\partial \gamma_p} = 0 = \frac{\partial E_{NOFT}}{\partial \gamma_p} - \mu \sin(2\gamma_p) \quad \rightarrow \text{Optimize } \mathbf{n} \text{ with l-BFGS}$$

Optimization of χ

Compute

$$\frac{\delta \mathcal{L}}{\delta \chi_p^*(\mathbf{r})} = 0 = \frac{\delta E_{NOFT}}{\delta \chi_p^*(\mathbf{r})} - \sum_r \lambda_{rp} \chi_r(\mathbf{r})$$

that multiplied by $\chi_q^*(\mathbf{r})$, and integrated over $\mathbf{r}$ leads to

$$\lambda_{qp} = \int d\mathbf{r} \frac{\delta E_{NOFT}}{\delta \chi_r^*(\mathbf{r})} \chi_q^*(\mathbf{r}),$$

The optimization II

We can obtain $\chi^{new} = \mathbf{U} \cdot \chi^{old} = e^{\kappa} \cdot \chi^{old}$ with κ being an anti-hermitian matrix.

Using the Adaptative Moments (ADAM) method (used in ML)

$$g_{pq} = 4(\lambda_{pq} - \lambda_{qp}) \quad \rightarrow p < q$$

$$\mathbf{m}^i = (\beta_1 \mathbf{m}^{i-1} + (1 - \beta_1) \mathbf{g}) / (1 - \beta_1^i)$$

$$v^i = (\beta_2 v^{i-1} + (1 - \beta_2) \|\mathbf{g}\|^2) / (1 - \beta_2^i)$$

$$v_{max}^i = \max(v_{max}^{i-1}, v^i)$$

$$\kappa_{pq} = \frac{\alpha m_{pq}^i}{\sqrt{v_{max}^i}}$$

where i accounts for the iteration, and β_1 , β_2 and α are parameters

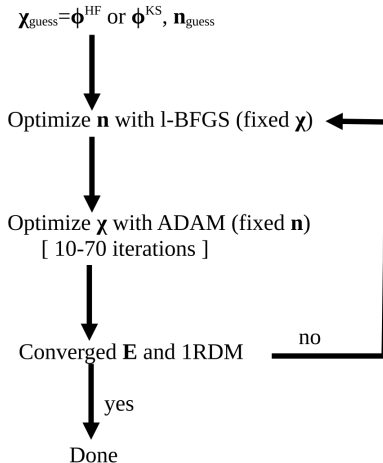
We obtain κ and $\mathbf{U}$ to optimize the natural orbitals!

J.F.H. Lew-Yee et al, *PRL*, **134**, 206401 (2025)

The optimization III

Algorithm

Is like doing HF a few times...



The DoNOF code

- ▶ Original DoNOF <https://github.com/DoNOF/DoNOFsw.git>



J.F.H. Lew-Yee et al, *JCP*, **164**, 072501 (2026)

- ▶ The standalone version
https://github.com/marm314/standalone_noft.git

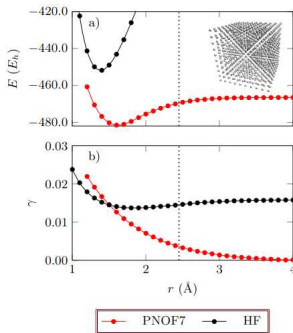


M. Rodriguez-Mayorga, *Zenodo*, (2022)

Results I

Energies and harmonic average (γ) for $10 \times 10 \times 10$ H atoms forming a cube (1s per atom)

→symmetric stretching of the cube



with

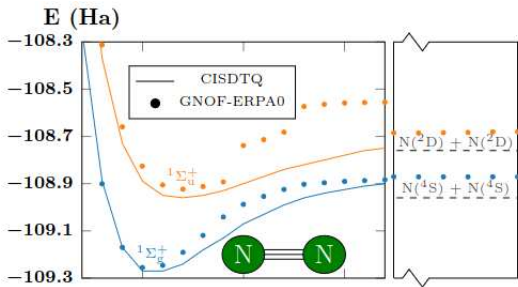
$$\gamma = \sqrt{\frac{1}{1000(1000 - 1)} \sum_{a \neq b} ({}^1D_{ab}^{1s})^2}$$

When $\gamma \rightarrow 0$, metal-to-insulator transition! (strong correlation)

Results II

Excited states with ERPax

Using an extension of RPax for multi-determinant Ψ (and cc-pVDZ basis set)



At the dissociated limit (strongly-corr. regime) the method performs well to describe the N atoms!

It is “not perfect” at the intermediate regime...

Other properties and extensions

- ▶ Total energies and (analytic) gradients → access to optimized lattice parameters, stability analysis, etc.
- ▶ Analytic Hessian (compute phonons)
- ▶ Extended for relativistic calculations (M. Rodríguez-Mayorga, *SciPost Chem.*, **1**, 004 (2022)) [deal with heavy elements]

SciPost

SciPost Chem. 1, 004 (2022)

Relativistic reduced density matrix functional theory

Mauricio Rodríguez-Mayorga*, Klaas J. H. Giesbertz and Lucas Visscher

Department of Theoretical Chemistry, Vrije Universiteit Amsterdam,
De Boelelaan 1083, 1081 HV Amsterdam, The Netherlands

* marm3.14@gmail.com

- ▶ DOS (S. Sharma et al, *PRL*, **110**, 116403 (2013))
- ▶ IP (EKT, S. Di Sabatino, *PRB*, **105**, 235123 (2022))

Within NOFT, the optimized 1RDM is correlated and the one-electron picture is lost...

Conclusions

- ▶ NOFT is an efficient path to describe strong-electronic correlation effects!
- ▶ Recent advances have facilitated its applicability to large systems
- ▶ Several properties can be extracted from NOFT calculations



Future perspectives

- ▶ Implementing NOFT in ABINIT!



Open to collaborations!

- ▶ Compute ph and $el - ph$ couplings (beyond the one-body picture!)...
RDMFT - PT (aiming to describe strongly-correlated systems).

Thank you!



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